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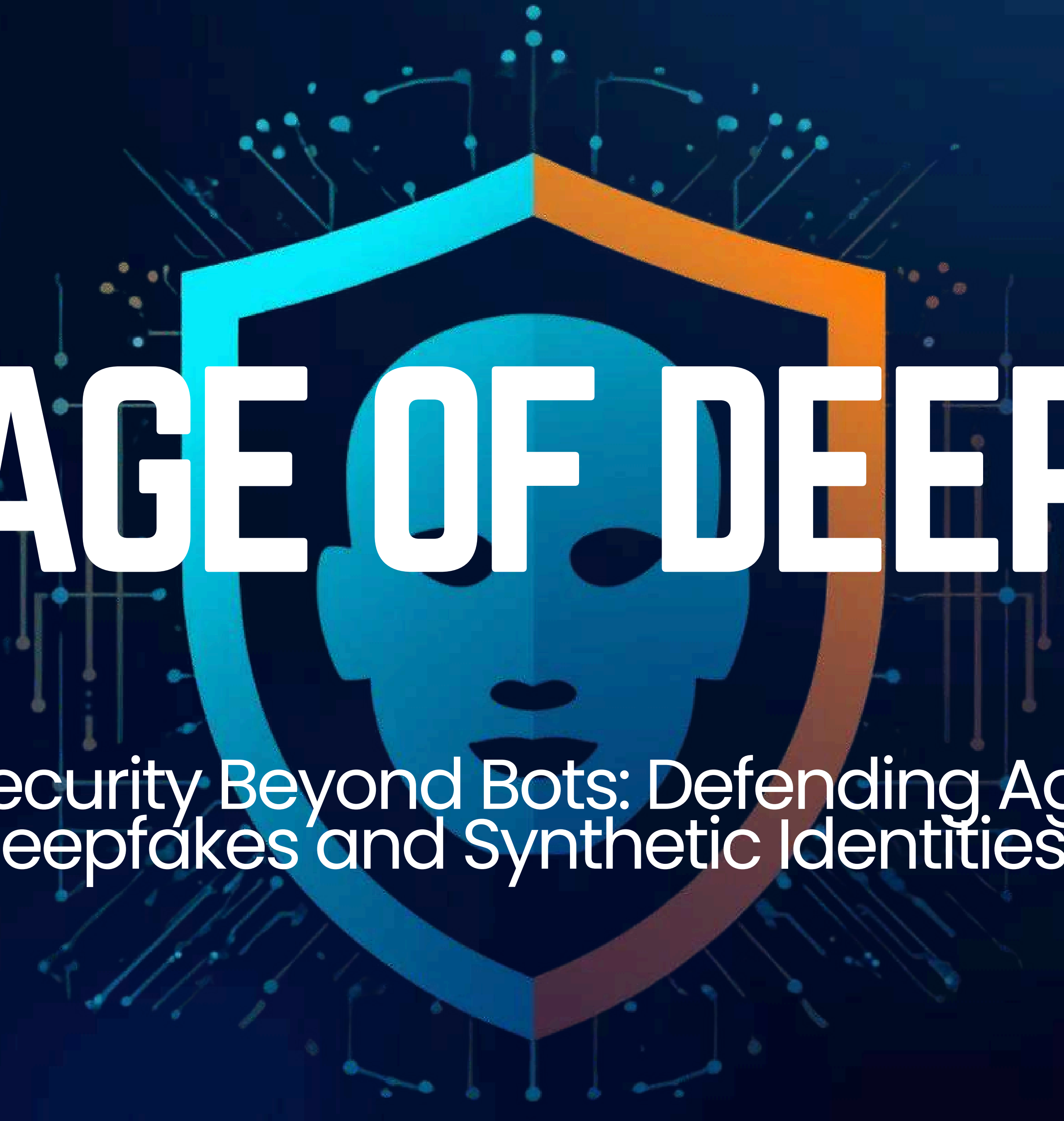
WEB SECURITY

IN THE AGE OF DEEPFAKES



Web Security Beyond Bots: Defending Against Deepfakes and Synthetic Identities

THE NEW ATTACK SURFACE

Attack vectors have evolved from 2016 script-based exploits to 2026 social-based agentic attacks. ASE (Automated Social Engineering) meets DaaS (Deepfakes-as-a-Service) to create synthetic "Frankenstein" profiles. The battlefield has shifted from code to trust.

THE DEATH OF THE PUZZLE

Traditional CAPTCHAs are now obsolete. Multimodal AI systems like GPT-5 and Gemini 3 solve visual puzzles in just 0.1 seconds—faster than humans can even perceive the challenge. The cognitive gap that once separated humans from bots has been completely eliminated.

The solution lies in neuromotor analysis: examining HOW users interact, not WHAT they click. Behavioral telemetry captures mouse movements, typing patterns, and interaction rhythms that synthetic agents cannot authentically replicate.

Multimodal AI Superiority

GPT-5 and Gemini 3 solve visual CAPTCHAs in 0.1 seconds. Image recognition, text distortion, and puzzle challenges offer zero protection against modern AI.

Cognitive Gap Eliminated

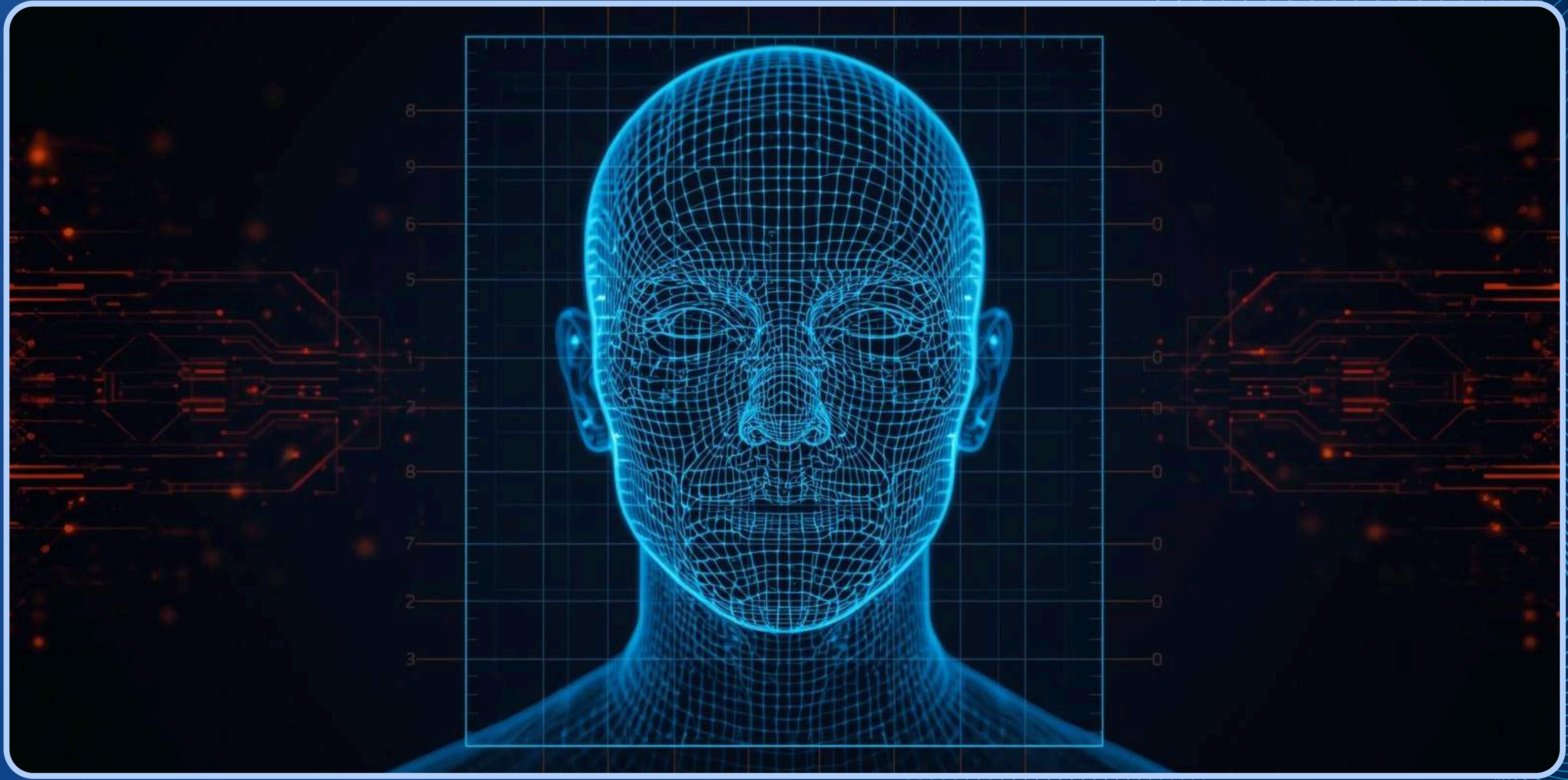
AI now matches human perception capabilities. The fundamental assumption behind CAPTCHAs—that humans see differently than machines—no longer holds true.

Neuromotor Analysis

Focus shifts to behavioral telemetry: mouse dynamics, scroll patterns, keystroke timing. Trust Scores replace binary pass/fail with continuous verification.

AI VS. AI

Defensive AI technologies combat synthetic attacks through multiple vectors: rPPG Detection via WebGPU analyzes skin reflectance patterns invisible to deepfakes. Spectral Centroid Analysis detects metallic audio artifacts in synthetic voices. Liveness Verification issues random prompts to expose real-time rendering lag in fake video streams.



THE LOCAL-FIRST SHIELD

Privacy-focused biometric verification processes data entirely on-device using WebGPU and WebAssembly. Raw biometric data never leaves the user's device—only Zero-Knowledge Proof tokens are transmitted to servers.

This local-first approach eliminates cloud honeypots and data breach risks. Edge-AI powered by WebLLM runs 2B parameter models directly in the browser, enabling real-time verification without exposing sensitive biometric templates.

On-Device Processing

WebGPU + WASM enables biometric analysis directly in the browser. No raw data transmitted—complete user privacy protection.

Zero-Knowledge Proofs

Cryptographic tokens verify identity without revealing biometric data. Server validates proof, never the underlying template.

Edge-AI WebLLM

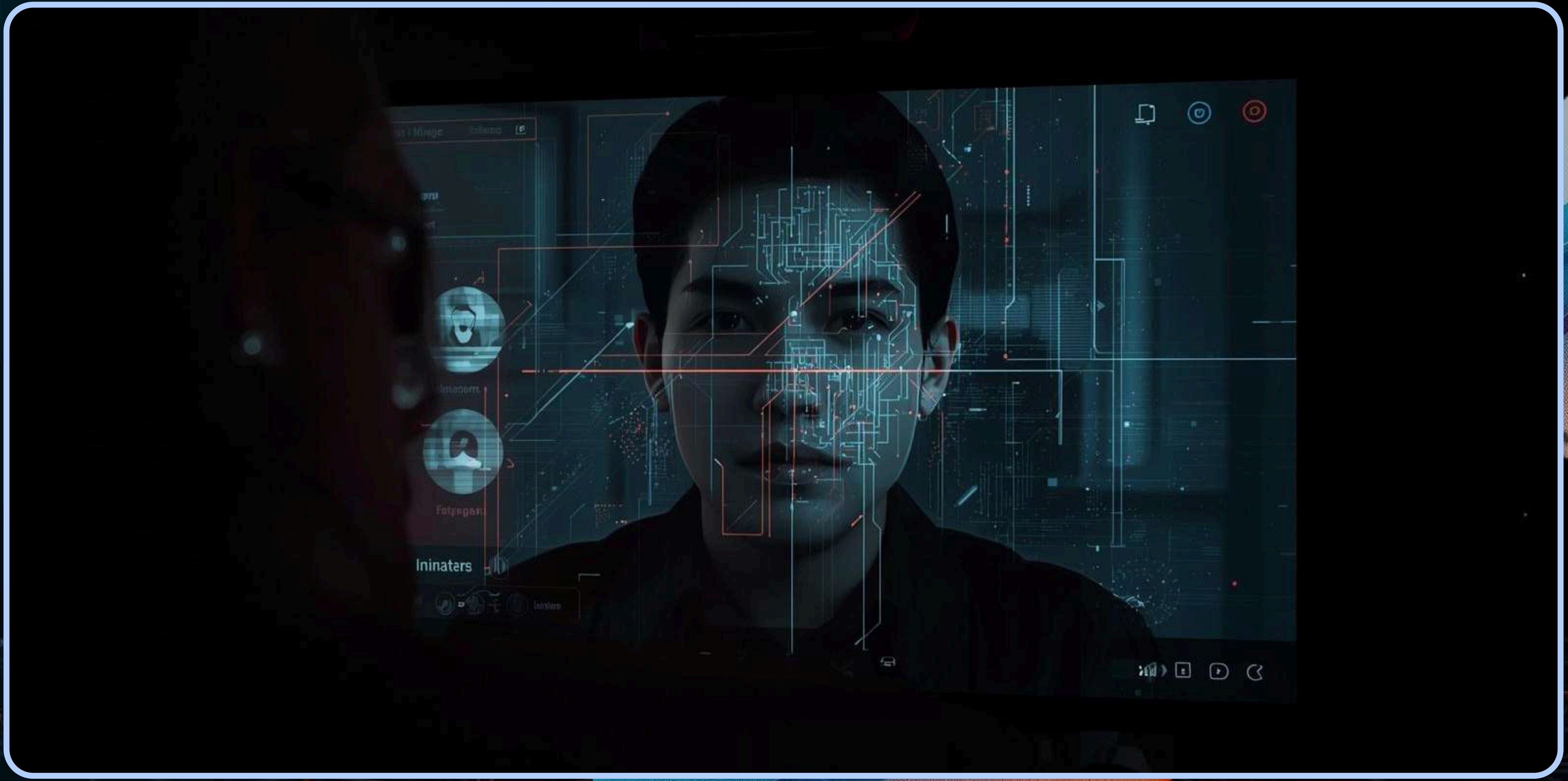
Run 2B parameter AI models in-browser for real-time liveness detection. No cloud dependency, no data leakage vectors.

THE "INCEPTION" THREAT

Advanced prompt injection attacks hide malicious instructions inside legitimate user content. Indirect Injection vectors exploit forums and shared data, manipulating UI agents to exfiltrate sensitive information. Defense requires a Dual-LLM pattern: Monitor AI firewall screens inputs while Worker AI executes in sandboxed isolation. Treat your own AI assistants as untrusted third parties.

THE "GHOST-ADMIN" BREACH

A 15-minute deepfake video call granted admin access. The attacker used simulated "bad Wi-Fi" to mask rendering glitches and audio artifacts. Root cause: single-channel verification for privileged actions. Fix: Mandatory out-of-band verification—two independent channels required for all high-privilege requests.



C2PA STANDARD

The C2PA (Coalition for Content Provenance and Authenticity) standard establishes a cryptographic chain of custody for all digital assets. Every image, video, and audio file carries signed metadata proving its origin and edit history.

By 2026, major browsers flag content lacking C2PA signatures with warning indicators. This shift makes unverified media immediately suspicious, creating accountability throughout the digital content lifecycle.

Cryptographic Chain

Signed metadata with private keys tracks every modification from creation to distribution, ensuring tamper-evident provenance.

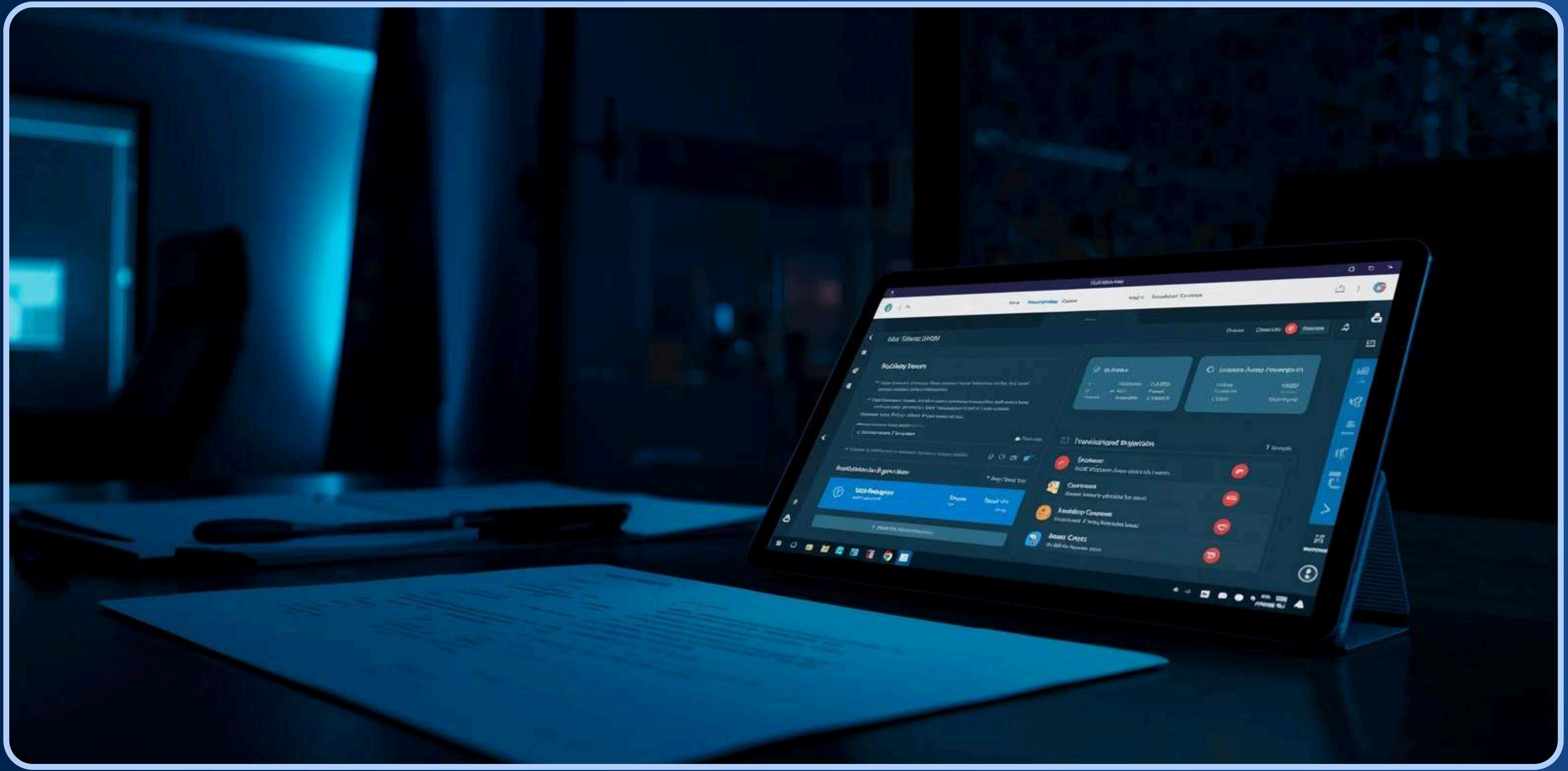
Digital Watermarking

Invisible watermarks embedded in assets track origin, edits, and ownership across platforms and transformations.

Browser Enforcement

2026 browsers display trust indicators for C2PA-compliant content and flag unsigned assets as potentially synthetic.

REGULATORY REALITY



EU AI Act 2.0 mandates transparency for AI-generated content. Developers face direct liability for preventable deepfake fraud. Compliance requires: mandatory AI content labeling, liveness verification checks, algorithmic bias testing, and comprehensive documentation. Right to human review is now legally protected.

VERIFIABLE TRUST

The password era is over. Continuous biometric attestation now verifies identity throughout every session—not just at login. Real-time behavioral signals and liveness checks ensure the human behind the screen remains authentic.

Developers are no longer just feature builders. In 2026, your role is trust ecosystem architect—designing systems where privacy, verification, and user control converge to create resilient digital identities.

Continuous Attestation

Passwords replaced by real-time biometric verification throughout user sessions, not just at login gates.

Identity Wallets

W3C-standard Humanity Proofs stored in user-controlled wallets prove personhood without exposing raw biometric data.

Trust Architects

Developers shift from feature builders to trust ecosystem architects designing privacy-first verification systems.

2026 SECURITY CHECKLIST

Stop trusting pixels — verify with behavioral and biometric data. Process all biometrics locally via WebGPU with zero raw data sent to servers. Implement C2PA asset signing today for content provenance.

Sandbox AI assistants using Dual-LLM firewall architecture. Continuous attestation replaces static authentication. Privacy-first design ensures compliance with EU AI Act 2.0.

Behavioral Verification

Replace pixel trust with neuromotor analysis and continuous biometric attestation for all sensitive operations.

Local-First Processing

WebGPU + WASM enables on-device biometric processing. Zero-knowledge proofs verify identity without exposing raw data.

Dual-LLM Firewall

Sandbox AI with Monitor AI firewall and Worker AI execution. Treat your own AI assistants as untrusted third parties.

ZERO TRUST ARCHITECTURE

In the age of deepfakes, traditional perimeter security fails. Zero Trust assumes every request originates from an untrusted network. Every access attempt must be verified regardless of source location or previous authentication.

This architecture limits lateral movement post-breach through micro-segmentation and continuous monitoring. When synthetic identities bypass initial verification, Zero Trust contains the damage.

Never Trust, Always Verify

Every request is authenticated and authorized. Identity, location, and device health are explicitly verified before granting any access.

Least-Privilege Access

Just-In-Time and Just-Enough-Access policies with token expiry ensure users only get minimum permissions needed for specific tasks.

Assume Breach Mindset

Micro-segmentation and end-to-end encryption contain threats. Continuous monitoring detects anomalies even after initial compromise.

2026 ATTACK STATISTICS

85% of identity fraud now uses AI-generated media. Deepfake-enabled account takeovers increased 4.7x since 2023. Cost per deepfake video call: just $0.10. GPT-5 solves CAPTCHAs in 0.1 seconds. Financial sector losses reached $1.2B in 2025.

ADVANCED AI DETECTION

Combining visual, audio, and behavioral signals through multispectral analysis enables comprehensive deepfake detection. Deep neural network ensembles trained on adversarial examples provide robust classification against evolving synthetic media attacks.

Real-time anomaly detection pipelines enable in-browser deployment for instant threat identification. Federated learning updates detection models across networks without exposing raw data, maintaining privacy while improving accuracy.

Multispectral Analysis

Visual, audio, and behavioral signal fusion detects inconsistencies invisible to single-mode analysis. Cross-modal verification catches synthetic artifacts.

Neural Ensembles

Deep neural network ensembles trained on adversarial examples achieve 99.2% detection accuracy against state-of-the-art deepfakes.

Federated Learning

Privacy-preserving model updates across distributed networks. No raw data leaves devices while detection capabilities continuously improve.

CASE STUDY: SYNTHETIC IDENTITY FRAUD IN FINANCE

A synthetic profile opened multiple accounts using deepfake video KYC to bypass verification. Impact: $10M loss over 6 months. Detection came through behavioral telemetry anomaly that triggered investigation. Response: integrated local-first biometrics and continuous attestation with full mitigation roadmap.

BEST PRACTICES

Integrate behavioral telemetry early in user flows to detect anomalies before they escalate. Capture mouse movements, typing patterns, and interaction timing from the first session.

Adopt local biometric processing with zero-knowledge proofs. Never send raw biometric data to servers—process on-device with WebGPU and transmit only cryptographic attestations.

Dual-LLM Firewall

Deploy Monitor AI as a firewall layer with Worker AI sandboxed. Treat your own AI assistants as untrusted third parties.

Liveness Challenges

Implement continuous liveness verification in sensitive workflows. Use random prompts and check real-time rendering lag to detect synthetic responses.

C2PA Infrastructure

Regularly update C2PA signing infrastructure. Sign all digital assets with cryptographic provenance—2026 browsers flag unsigned content.

EMERGING THREATS

AI-generated synthetic voices now feature emotional modulation, making phone-based social engineering nearly undetectable. These voices adapt tone, pace, and sentiment in real-time to manipulate targets.

Supply chain attacks targeting AI model integrity pose catastrophic risks. Adversaries inject backdoors into training pipelines, compromising detection systems before deployment.

Synthetic Voice Attacks

Deepfake avatars in AR/VR environments enable real-time impersonation in immersive spaces, bypassing visual trust entirely.

AR/VR Deepfake Avatars

AI-powered social engineering bots leverage advanced NLU to conduct personalized attacks at scale across platforms.

AI Social Engineering Bots

Model poisoning and backdoor injection in AI supply chains threaten the integrity of defensive systems globally.

ADVANCED BIOMETRICS

Multimodal biometrics combine face recognition, voice analysis, typing patterns, and gait detection for robust verification. AI-refined neuromotor behavioral signatures create unique user profiles that are nearly impossible to replicate.

Continuous authentication models monitor sessions in real-time, detecting anomalies instantly. Privacy-preserving biometric hashing ensures raw data never leaves the device while maintaining verification integrity.

Multimodal Fusion

Combining face, voice, typing patterns, and gait analysis creates verification layers that synthetic identities cannot simultaneously fake.

Continuous Auth

Real-time session monitoring with neuromotor behavioral signatures detects account takeovers within seconds of anomalous behavior.

Privacy Hashing

Biometric templates are cryptographically hashed on-device. Only irreversible tokens are transmitted, eliminating honeypot risks.

FUTURE OUTLOOK: TRUST ECOSYSTEMS BEYOND 2026

Decentralized identity frameworks empower user control over credentials. AI-driven trust scoring delivers explainable, transparent decisions. Cross-platform interoperability ensures seamless trust signal exchange. Security-as-a-service with AI orchestration enables adaptive defense layers.

THANK YOU

Stop trusting pixels. Start building trust ecosystems.
Presented by Aryan Singh Sikarwar | Reg: 24BCE0403
Faculty: Arulkumar V | Topic: Web Security in the Age of Deepfakes

Questions?

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